

Mohammed Thaher S

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SUMMARY

CS graduate who builds and ships production-ready full-stack and AI-integrated systems. Experienced in designing custom Hybrid ReAct architectures, RAG pipelines, and end-to-end multimodal applications. Handled frontend-to-backend integration for the TravelKeet platform launch at **IIT Madras Research Park**. Integrates LLM agents and workflows directly into products, served via FastAPI and Django, with live deployments across multiple cloud platforms. Proven track record in national hackathons with a focus on delivering working, high-accuracy systems.

SKILLS

Languages & Frontend: Python, JavaScript, HTML, CSS, React, Vite

Backend & APIs: Django, FastAPI, Flask, REST APIs, PostgreSQL, MySQL, MongoDB

AI & ML: LangChain, LangGraph, FAISS, Pinecone, YOLOv8, OpenCV, PyTorch, Scikit-learn, Hugging Face, Groq, OpenAI, Gemini

Tools & Deployment: Git, GitHub, Docker, Streamlit, Gradio, Postman, Pytest, CI/CD basics

EXPERIENCE

Campervan Factory, IIT Madras Research Park (IITMRP)

Jun 2025 – Sep 2025

Backend Developer Intern

Chennai, India

- Integrated frontend UI components with a Django backend for the TravelKeet platform launch, ensuring smooth data flow and session management across 30+ views
- Collaborated within a cross-functional team in an active codebase, using Git for version control, feature branching, and coordinated delivery
- Managed and mentored 2 fellow interns, coordinating tasks and establishing standard branching conventions for the team

PROJECTS

Trendly – Hybrid ReAct Support Agent | *Groq (Llama3), FastAPI, Pytest, Custom ReAct Engine* | [Live Demo](#)

Jul 2026

- Built a **custom Hybrid ReAct architecture**: Python-native ReAct loop governing a Groq LLM, backed by a **deterministic rules engine with 8 business-logic guardrails**
- Served via async **FastAPI** with Pydantic request validation; implemented Pytest suites covering ID normalization, blocked categories, final-sale constraints, and delay credit eligibility

DocuMind – Autonomous Document-Generation Agent | *LangGraph, FastAPI, Groq, React, Vite* | [Live Demo](#)

Sep 2026

- Engineered a **LangGraph state machine with 5 nodes** (plan, execute, reflect, revise, export) that classifies 7 document types, generates sections, and runs a self-critique pass before exporting a polished .docx
- Full-stack deployment: React frontend served from the same async **FastAPI** process with Pydantic validation; deployed on Railway

Multimodal Visual Q&A System | *FastAPI, YOLOv8m, Gemini 1.5 Flash, Gradio, Docker* | [Live Demo](#)

May 2026

- End-to-end deployed **FastAPI** backend with modular routers separating YOLOv8m CV inference (80 COCO classes) from Gemini 1.5 Flash LLM reasoning as independently replaceable services
- Containerized with **Docker**; Gradio frontend deployed on Hugging Face Spaces using async I/O for non-blocking concurrent requests

HACKATHONS & RECOGNITIONS

AgriTech Hackathon – Winners | *Pavaman Technologies* | [Live Demo](#)

May 2025

- Fine-tuned **EfficientNetB4** for 38-class crop disease detection; achieved **95% classification accuracy** and shipped a live Streamlit demo under competition deadline

Zelestra x AWS ML Hackathon – Finalist | *HackerEarth* | [Live Demo](#)

Oct 2025

- Built a stacking ensemble (CatBoost, LightGBM, TabNet, XGBoost meta-regressor) with Optuna HPO over **50 trials**; achieved **94.14% evaluation score** with SHAP interpretation

EDUCATION

Sathyabama Institute of Science and Technology

Jun 2022 – Jul 2026

B.E. Computer Science and Engineering | First Class with Distinction

Chennai, India

CERTIFICATIONS

Oracle Cloud Infrastructure Associate | Intro to MongoDB (*MongoDB University*)